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清華大學 Tsinghua University

Aerodynamic Optimization using CFD for a Hybrid eVTOL in Cruise

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Introduction

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Thesis overview

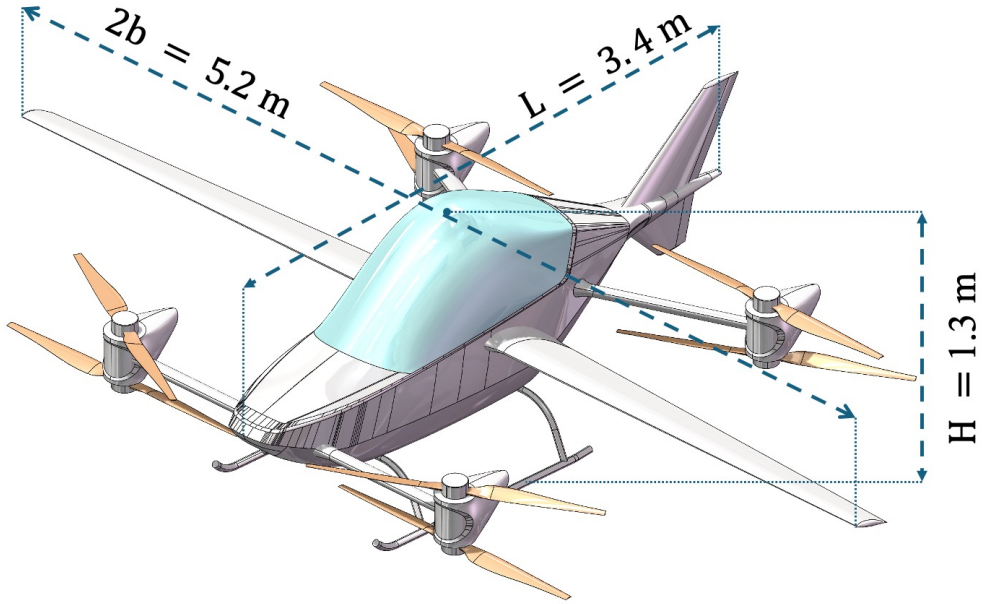
Motivation

- Cruise efficiency is strongly penalized by **rotor-wing interference** in hybrid quad-rotor eVTOLs

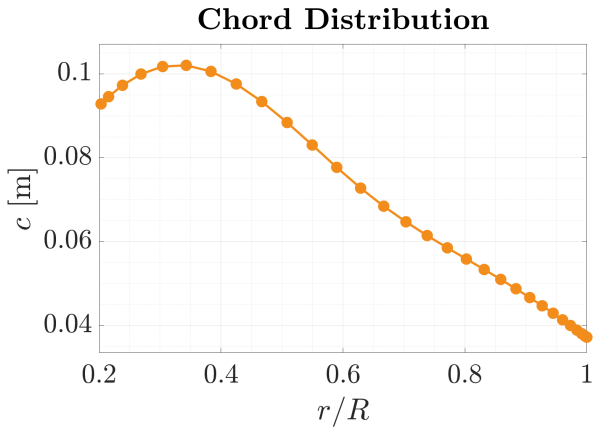
Objective

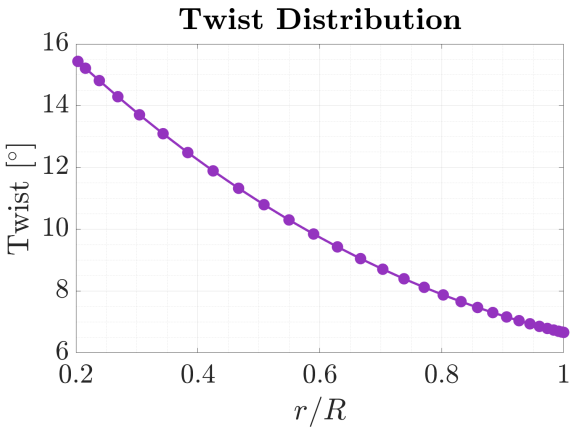
- **Power** consumption **reduction** in cruise

Target Vehicle – the W90









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Setup and Case Analysis

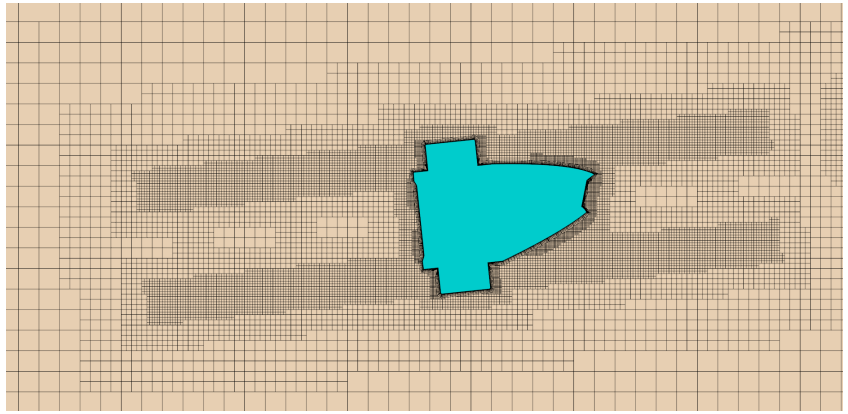
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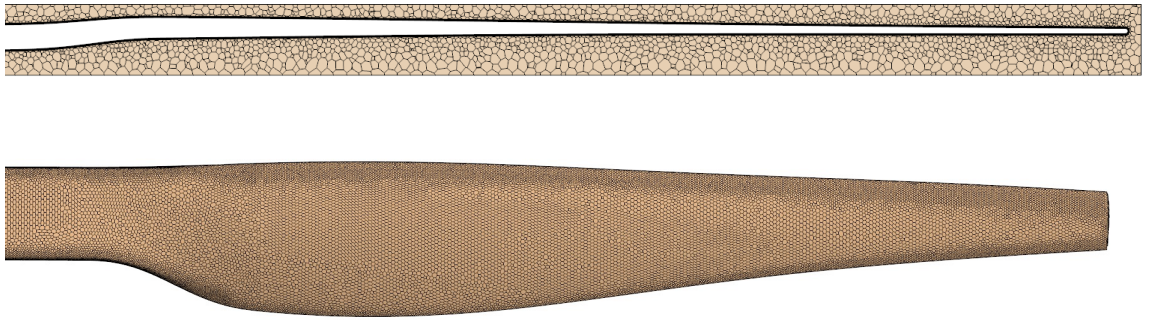
Mesh topology

VDM



- **Trimmed-cell** mesh
- Local refinements

SMM



- VDM mesh far from rotors
- **Polyhedral** mesh in the rotating domains

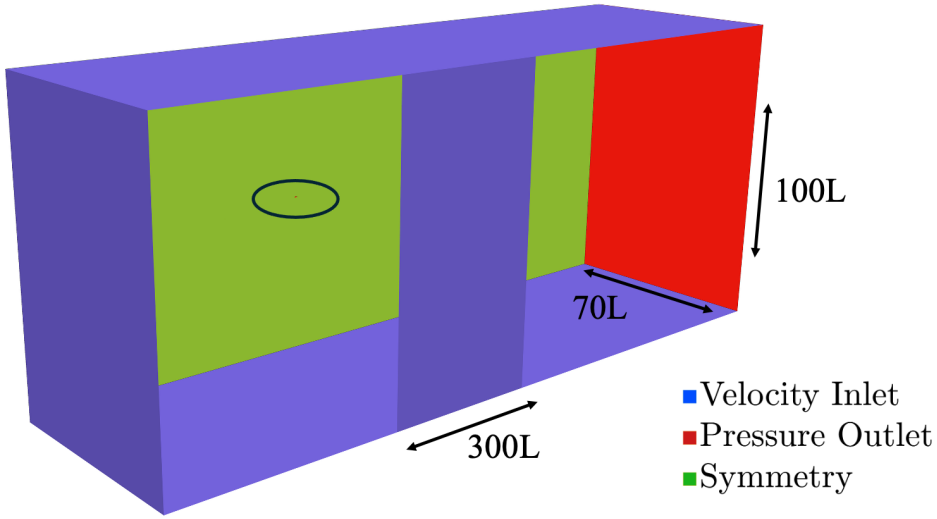
Solver setup

VDM

- **Incompressible** flow ($V_\infty = 30m/s$)
- **Segregated** flow
- $k - \varepsilon$
- **Steady**-state RANS
- Symmetry
- Wall functions

SMM

- **Compressible** flow ($V_\infty = 30m/s$)
- **Coupled** flow
- $k - \omega$ SST
- **Unsteady** RANS
- Symmetry
- Wall functions

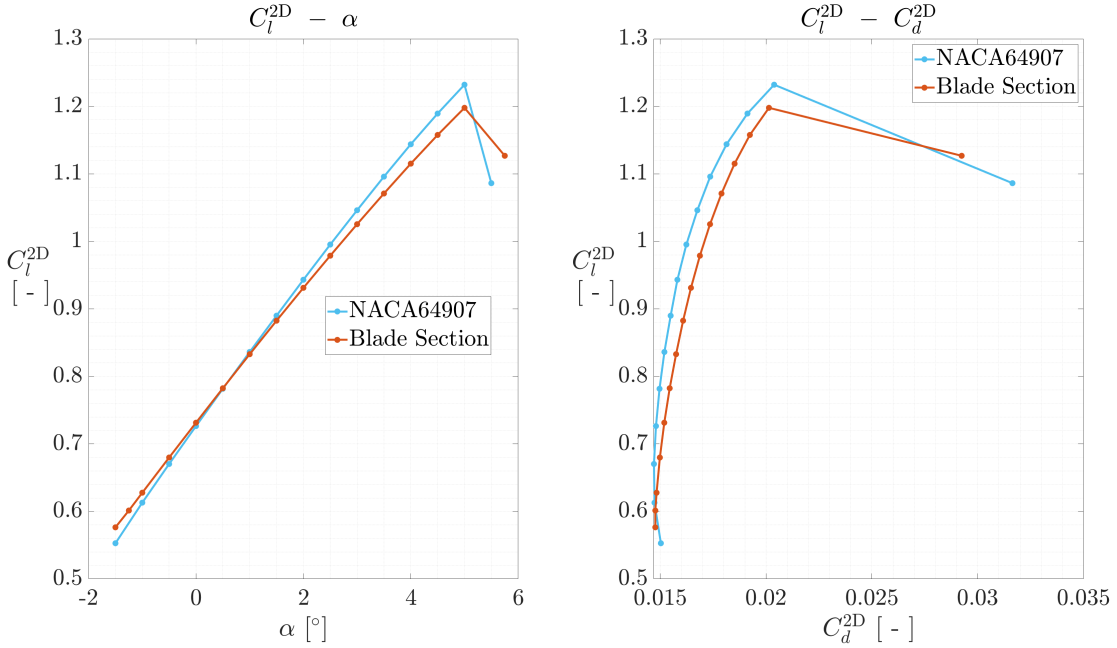


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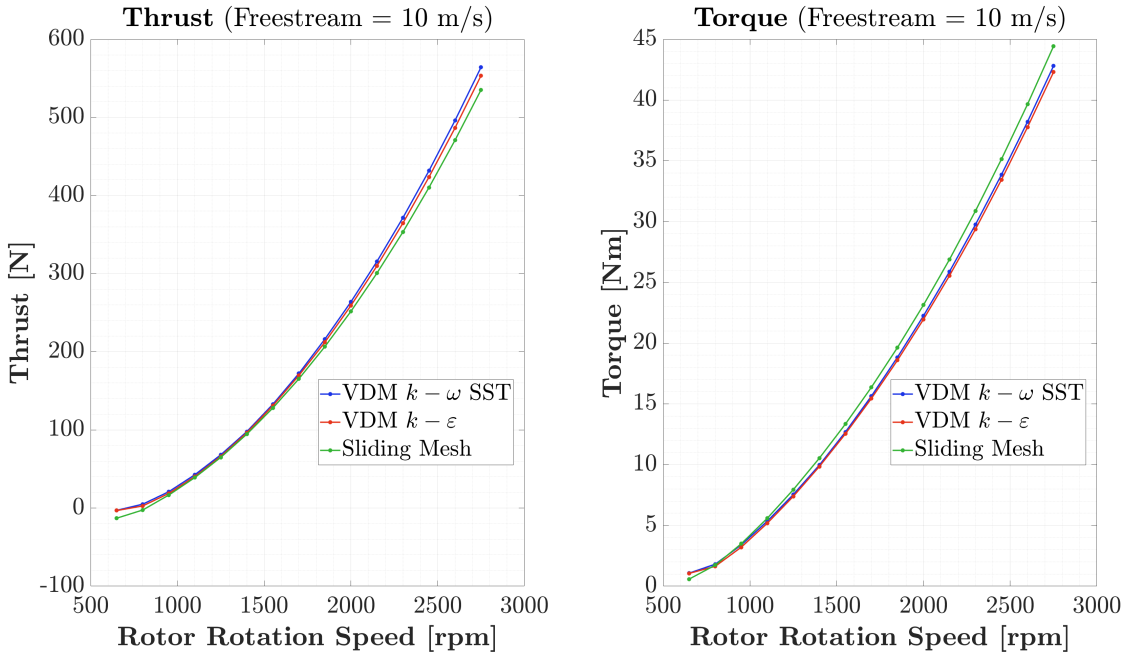
Validation – VDM

Actual blade *vs.* NACA64907 2D polars



- Linear region

VDM *vs.* validated SMM results (axial flow)



- Operative conditions:

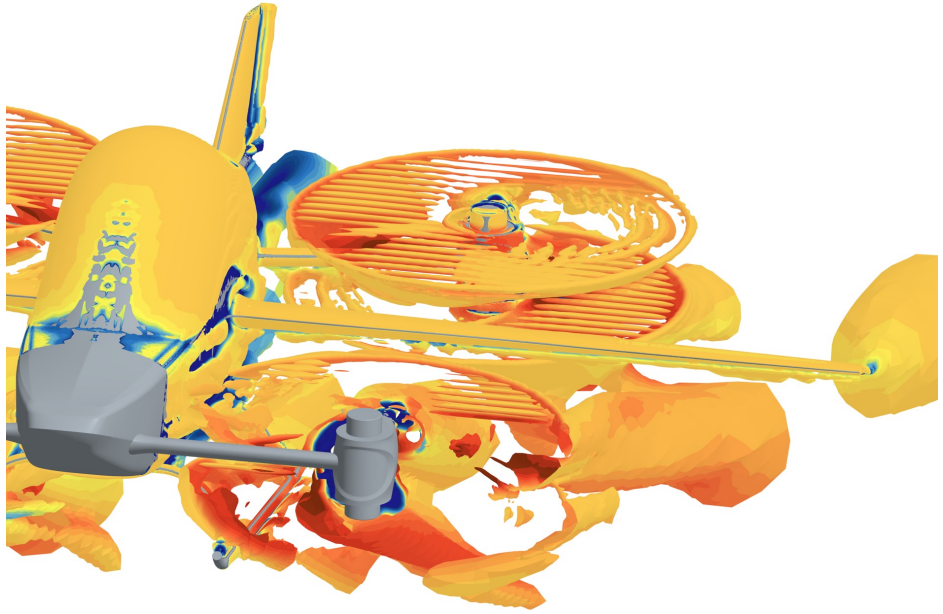
$\Omega \sim 1300\ rpm$

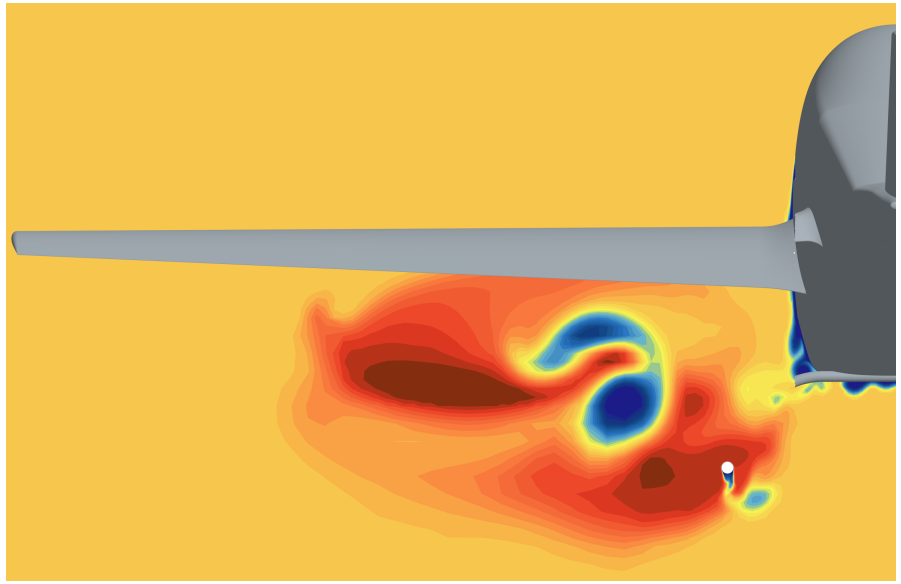
$V_{\perp} \sim 10 m/s$

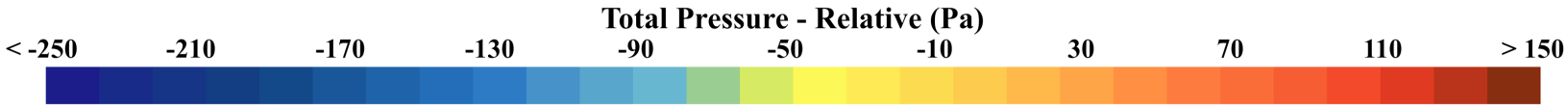
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Case analysis (1)

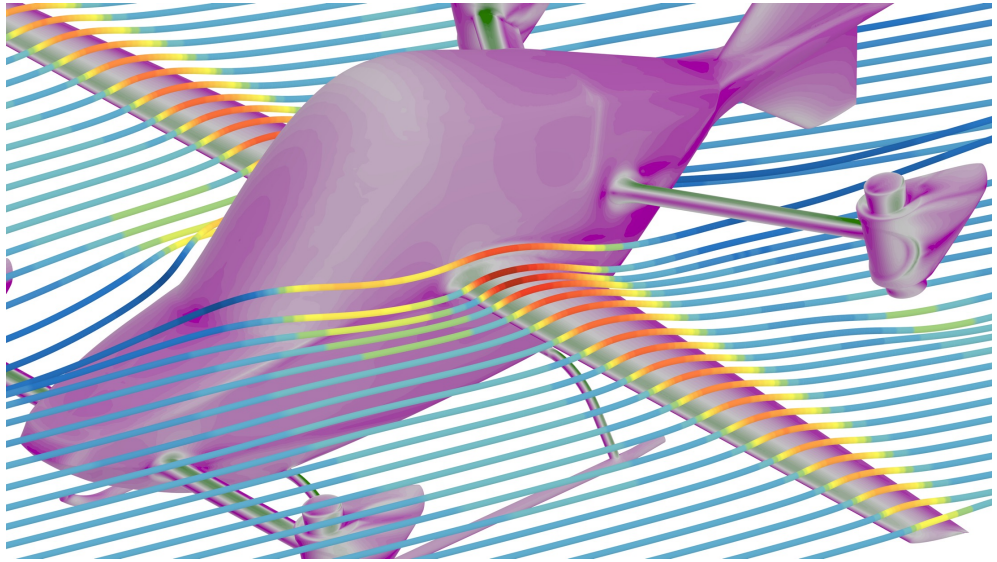




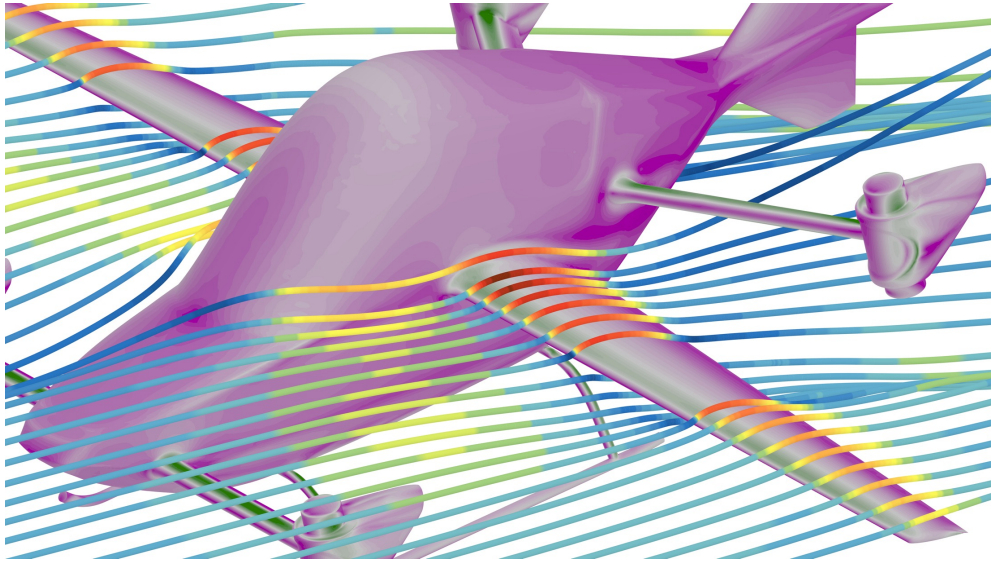


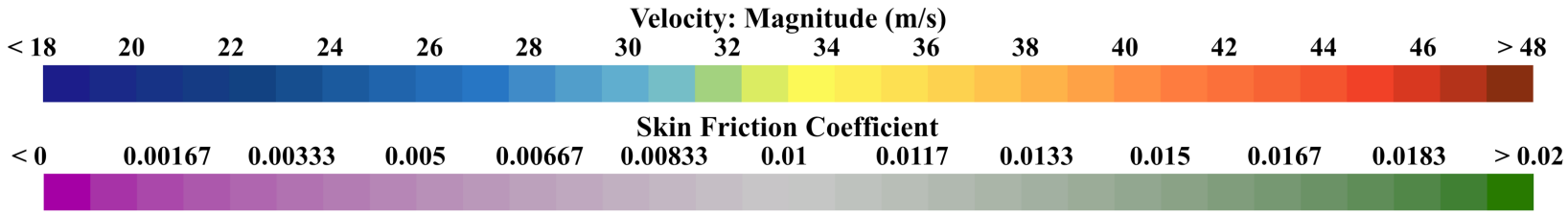
Case analysis (2)

Rotors-off

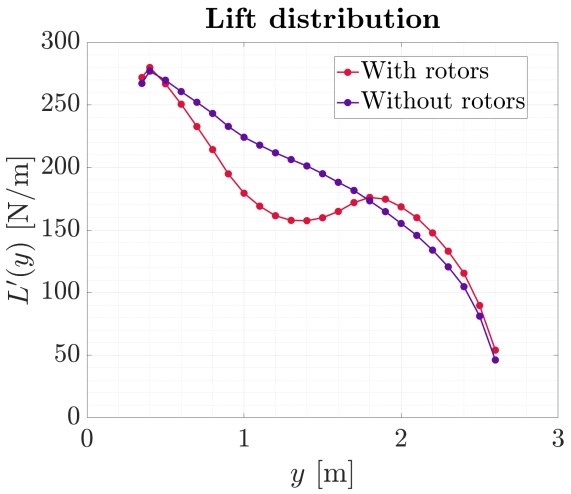


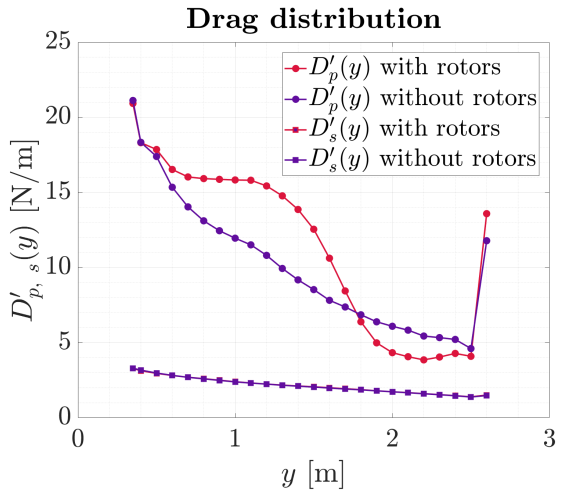
Rotors-on

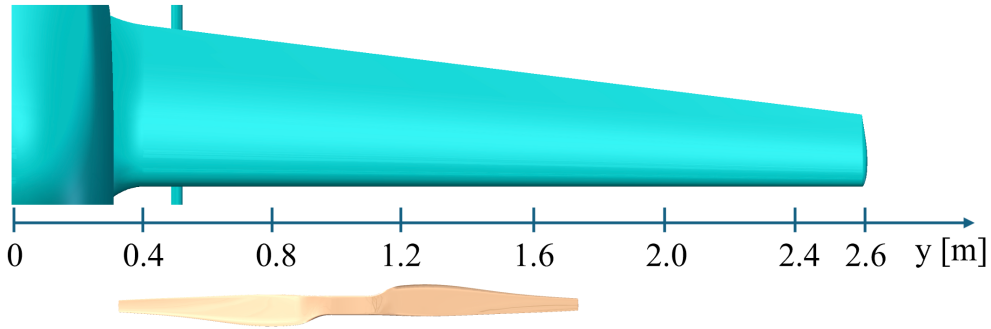


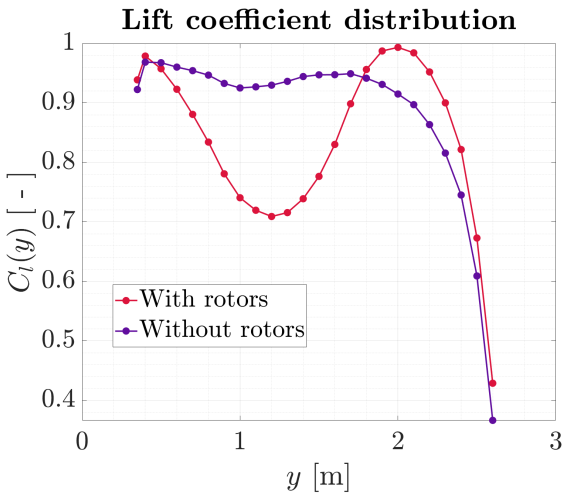


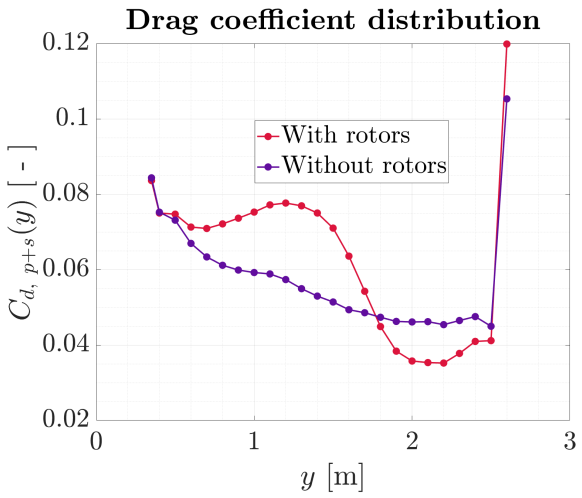
Case analysis (3)











- **Downwash** inboard
- **Upwash** outboard
- Peak at $y \sim 1.2\ m$
- **Performance degradation**

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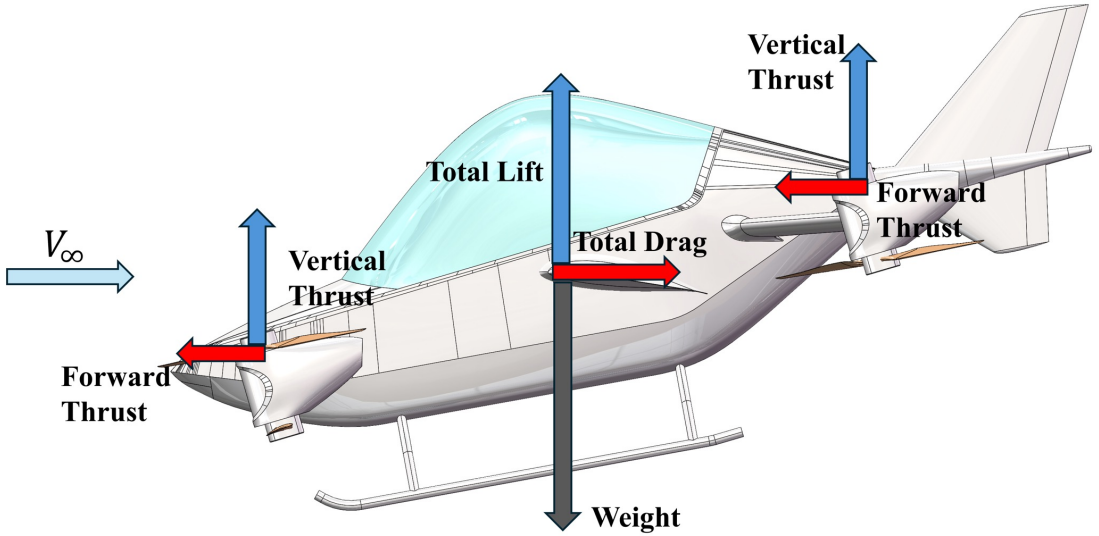
Design Strategies and Performance Evaluation

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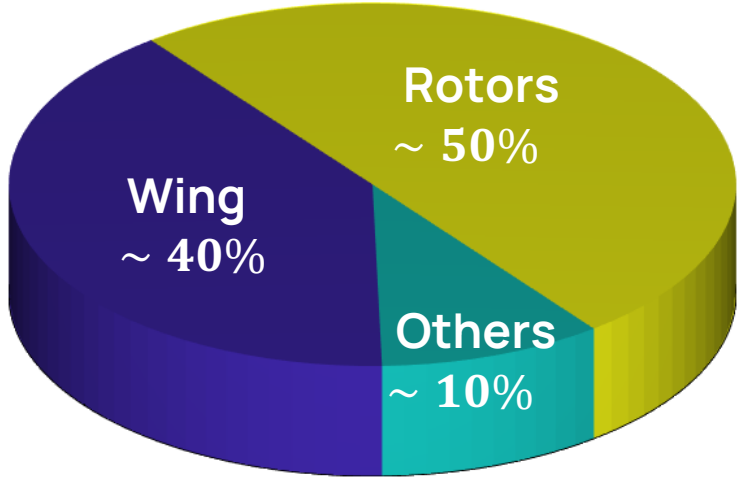
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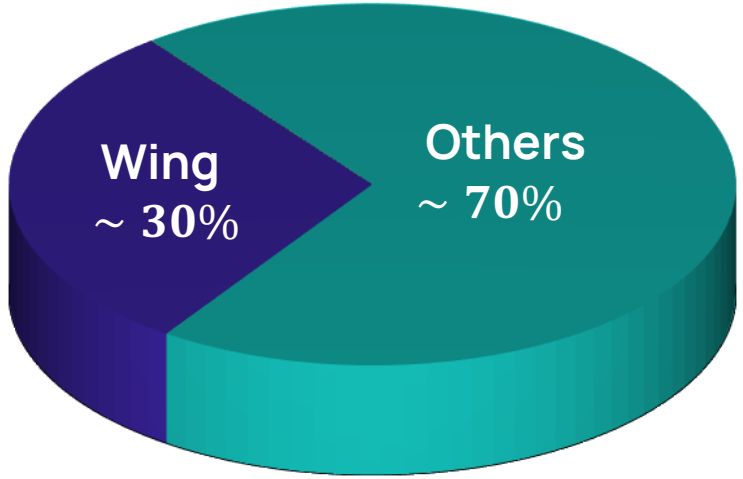
Optimization environment – heuristic approaches



Lift shares



Drag shares



Strategies:

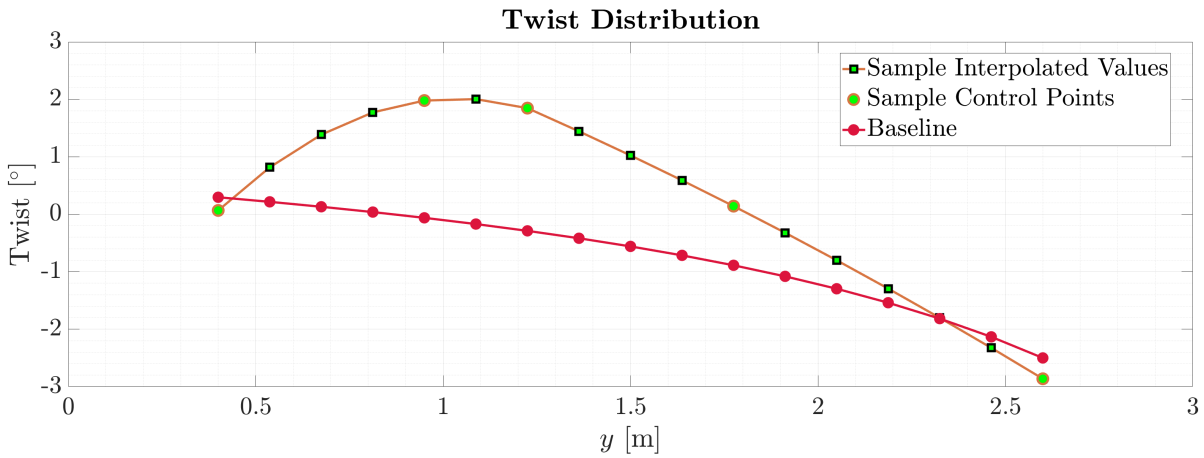
- Lift-to-drag ratio increase
- Lift increase

Optimization environment – setup

Optimization Setup	
Objective	$\max L^w/D^w$
Variables	Twist at $y \sim [0.4, 1, 1.2, 1.8, 2.6]$ m
Constraints	Spanwise twist gradient ≤ 3 °/m $L^w \geq L_{\text{ref}} = 408$ N $C_l(y) \leq \bar{C}^{2\text{D}}_{l,\,\max} = 1.15$ $C_l(y) \leq 1.15\ C_l(y = 0.4)$
Algorithm	Genetic Algorithm
Evaluation	Surrogate-based (Kriging)

Bounds	
Twist at y ∼	**[min, max]** [°]
0.4 m	[-2.5, 1.5]
1 m	[-2.0, 3.5]
1.2 m	[-2.5, 3.0]
1.8 m	[-3.5, 2.0]
2.6 m	[-4.0, 0.5]

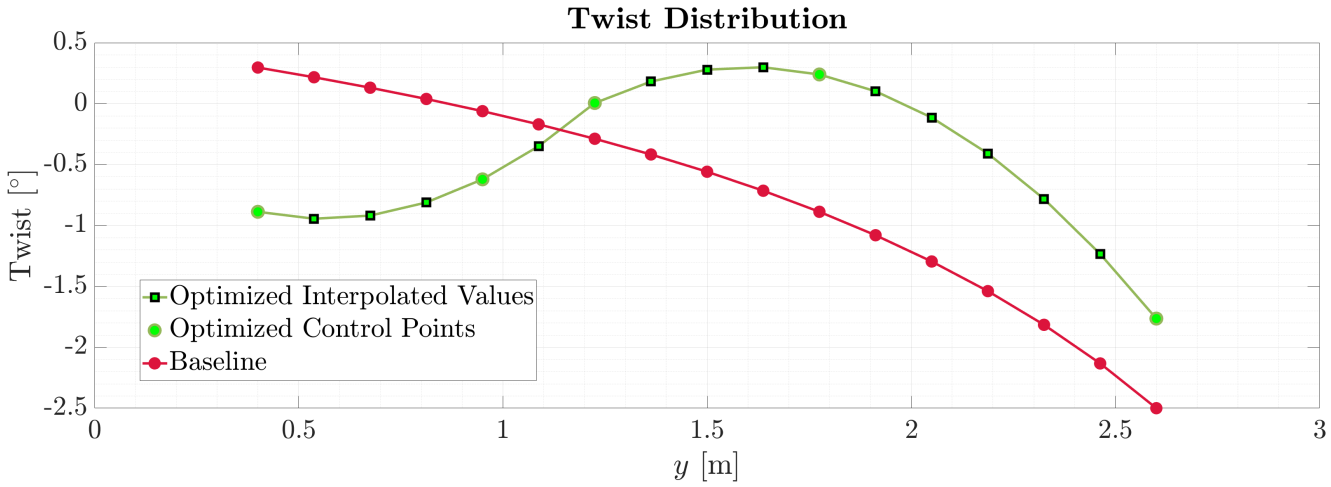
- Structural feasibility
- Minimum lift
- Stall margin (critical section theory)
- Tip loading

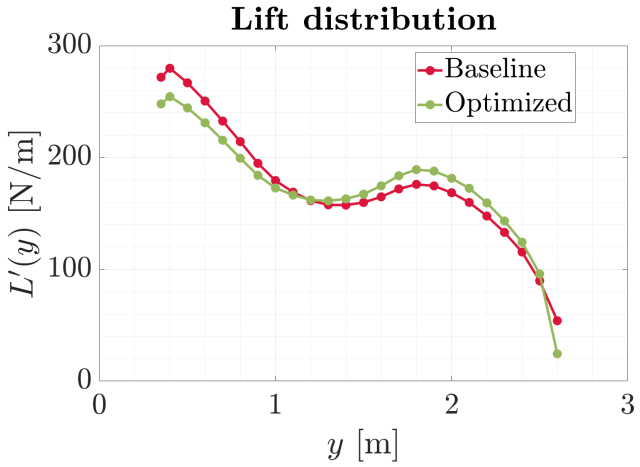


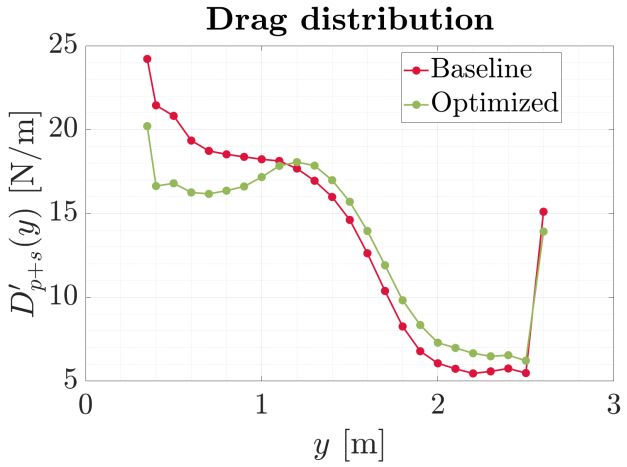
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Twist optimization results







- **Wash-in** inboard, **wash-out** outboard
- **Constraints** compliance

$$\max\left(C_l(y)\right) = 1.07$$

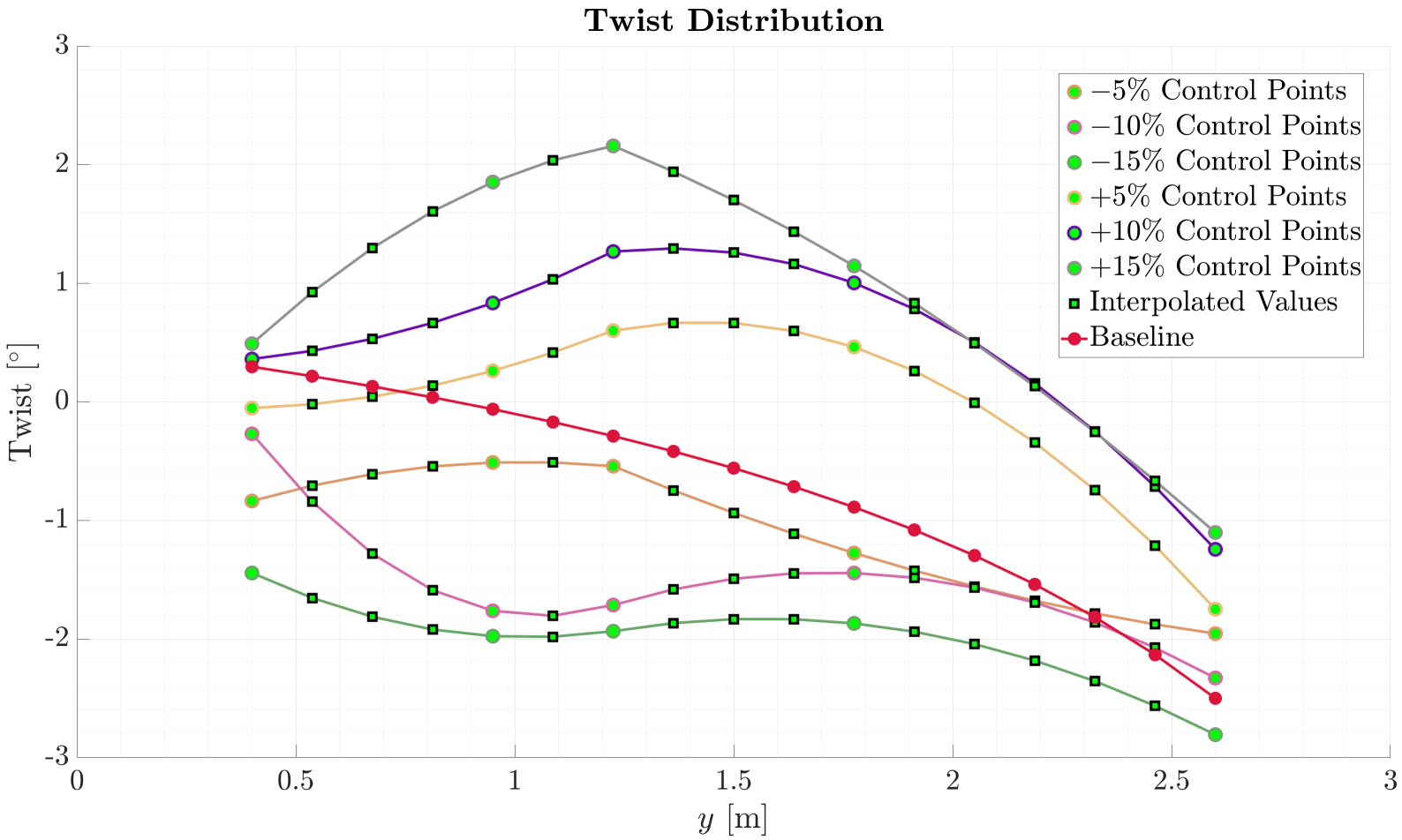
- 2.4% improvement on L^w/D^w → **negligible** effects on P

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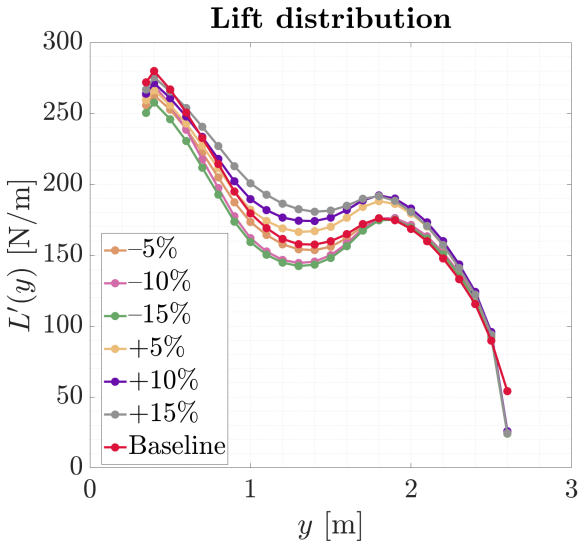
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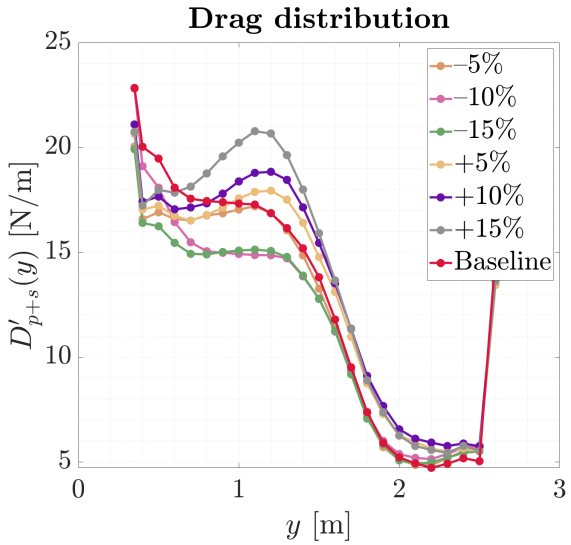
Lift-based design (1) – strategy and twist distributions

$$\mathbf{L}_{\mathbf{ref}} = (\mathbf{1} \pm [5\%, 10\%, 15\%])L_{\mathbf{base}}, \qquad L_{\mathbf{base}} = 408 \text{ N}.$$

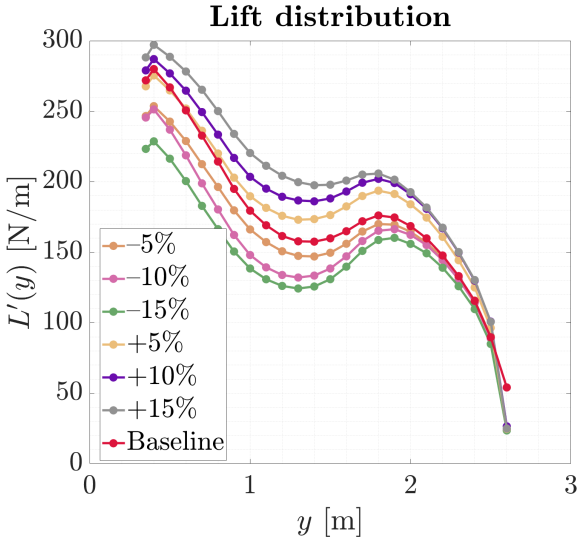


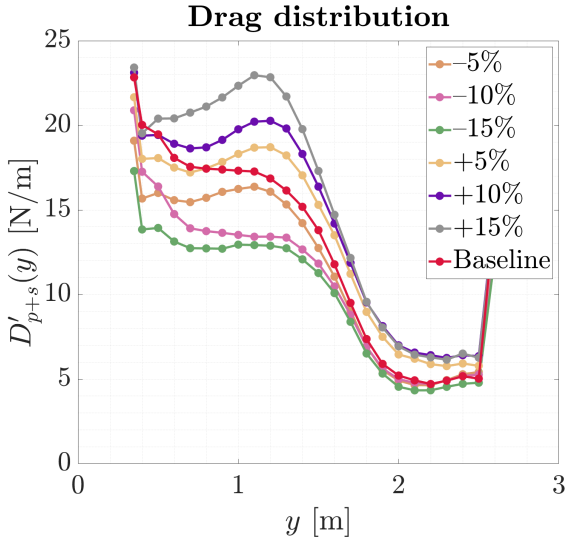
Lift-based design (2) – trim strategies and load distributions





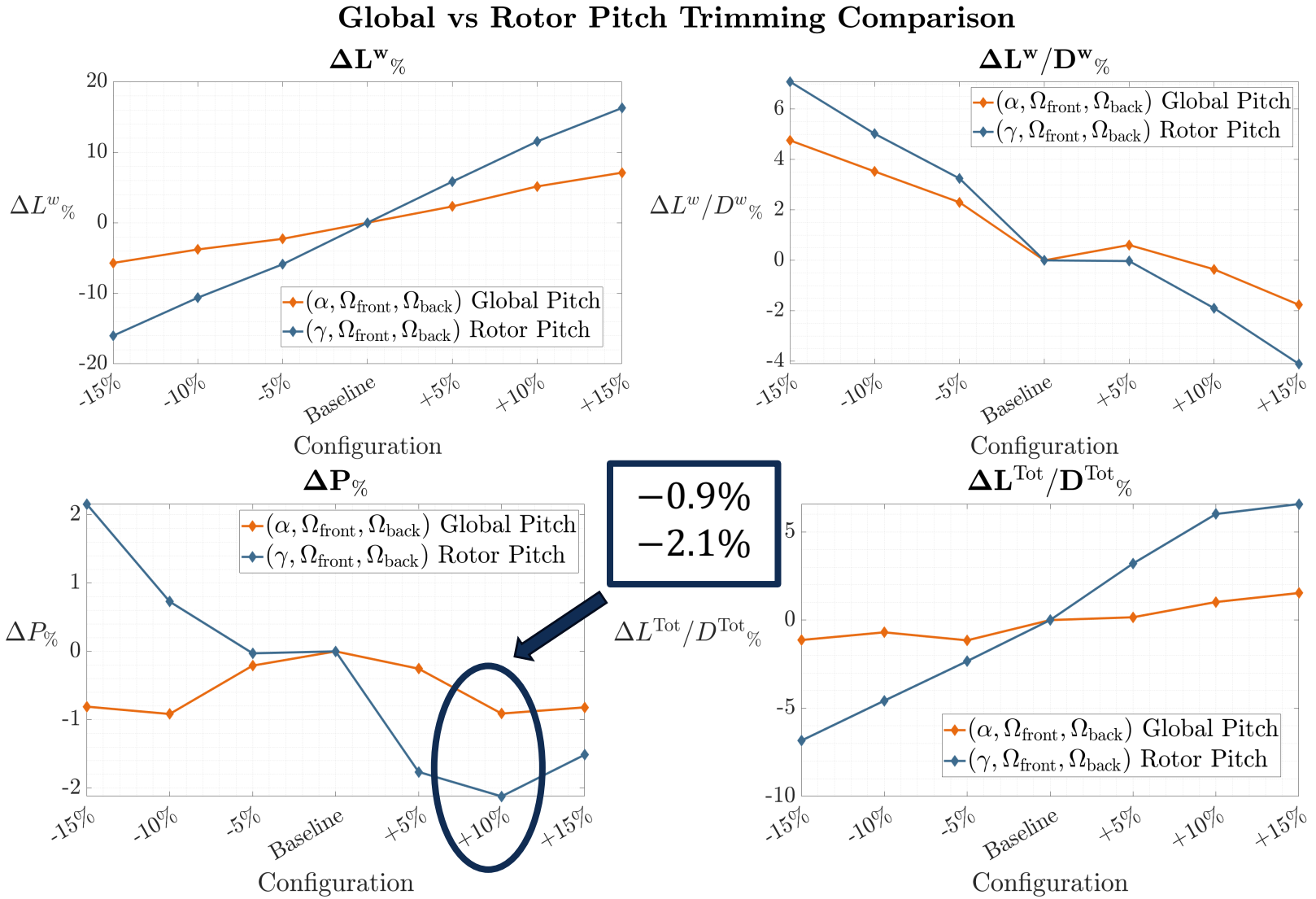
$(\alpha, \Omega_{\text{front}}, \Omega_{\text{back}}) \rightarrow$ **Global** pitch trimming





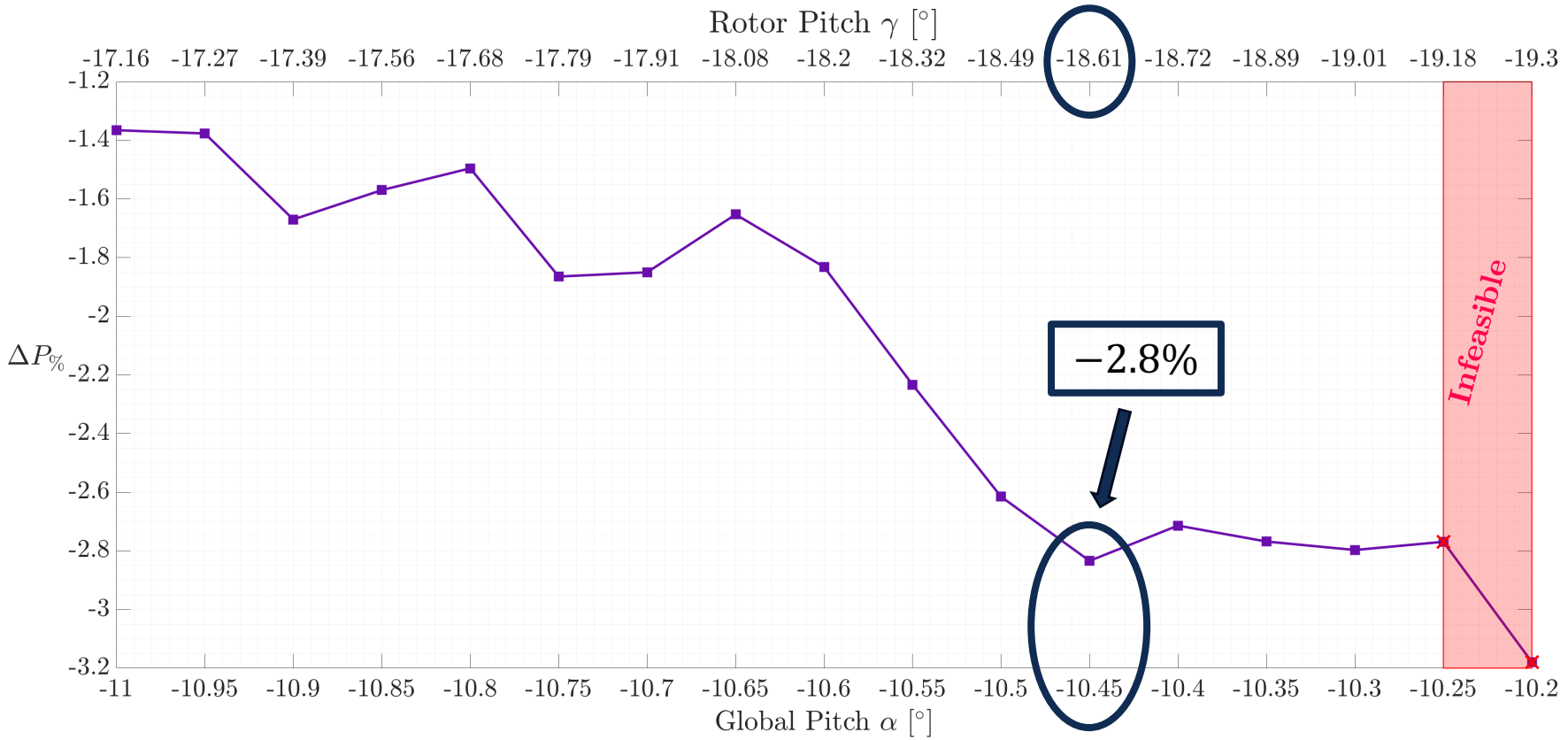
$(\gamma, \Omega_{\text{front}}, \Omega_{\text{back}}) \rightarrow$ **Rotor** pitch trimming

Lift-based design (3) – integral results



Lift-based design (4) – four-parameter trim strategy

$(\alpha, \gamma, \Omega_{\text{front}}, \Omega_{\text{back}})$ → Four-parameters trimming



Baseline:

$\alpha = -10.59°$

$\gamma = -16.19°$

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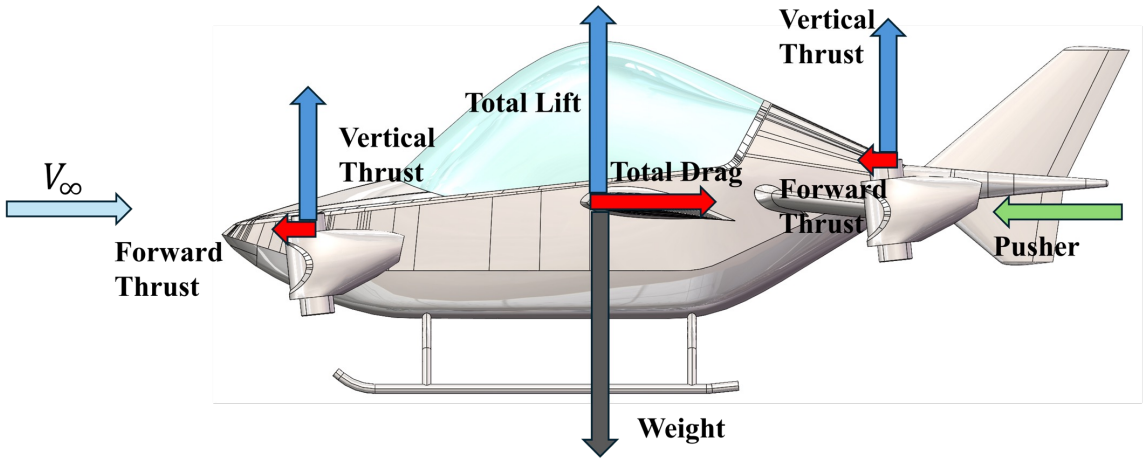
Alternative design (1) – strategy overview



Cento [1]



Pipistrel [2]



- **Dual-system** configuration
- Wing incidence modification

Alternative design (2) – results

$$P_{\text{pusher}} = \frac{T_{\text{pusher}} V_\infty}{\eta_p} \qquad P_{\text{rotors+pusher}} = P_{\text{rotors}} + P_{\text{pusher}}$$

✓ Aerodynamic performance **increase**

✗ Power consumption **increase**

} Why? → Model uncertainty assessment

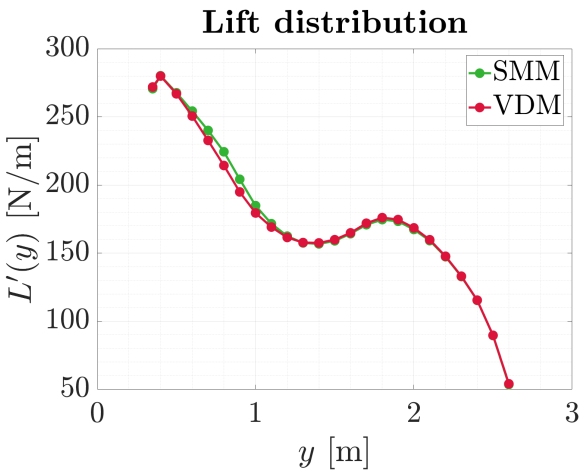
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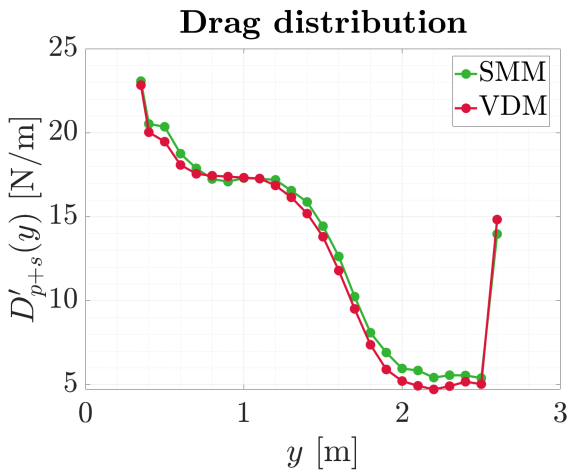
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Model Uncertainty Assessment

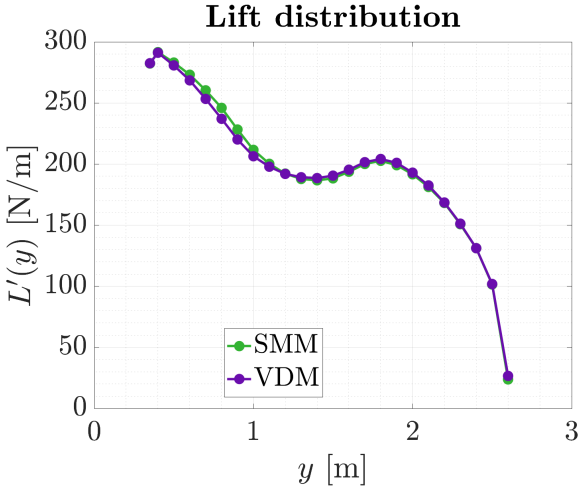
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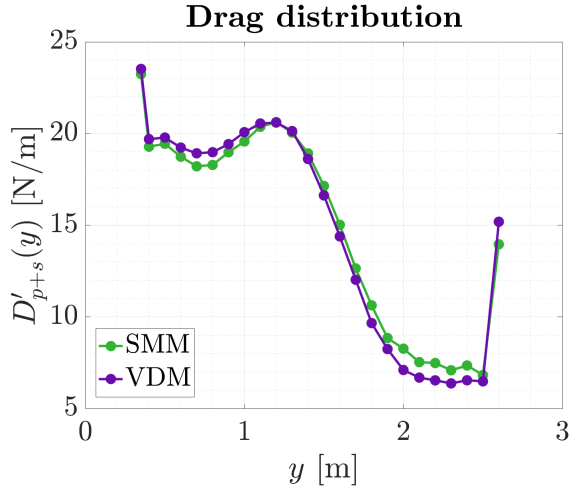
Model uncertainty assessment (1) – baseline and +10% configurations





Baseline





+10%

Vehicle aerodynamic loads:

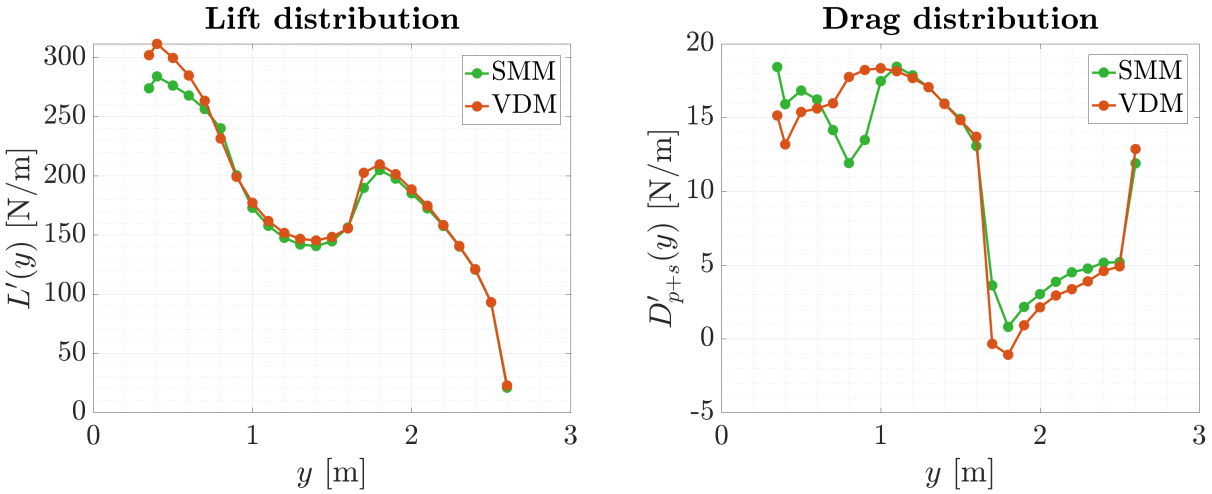
- Good agreement locally
- Up to ~ 6% discrepancy on integral loads

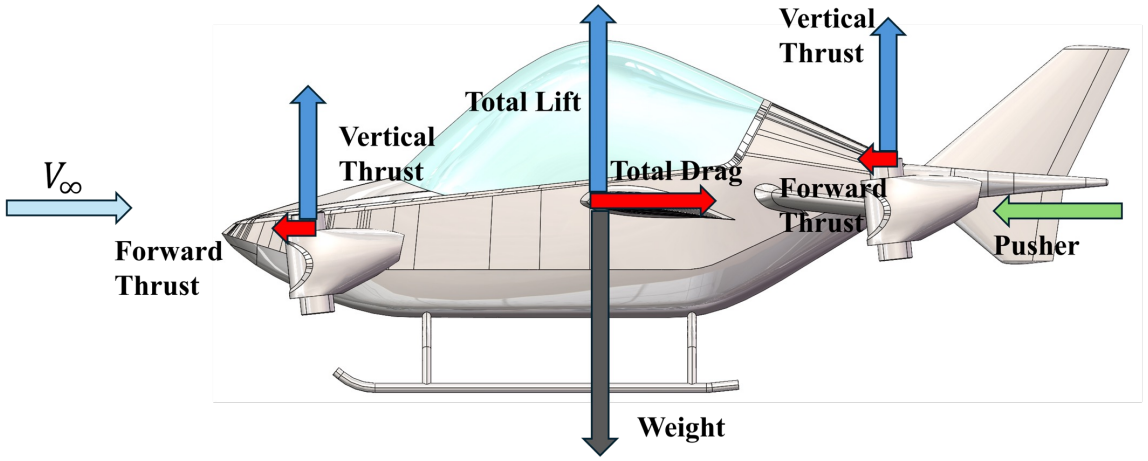
Rotor loads:

- More than 10% variation in thrust
- 15– 25% deviations in torque, power consumption
- Up to 1.7° deviations in thrust tilt angle

Model uncertainty assessment (2) – dual-system configuration

Dual-system





Vehicle aerodynamic loads:

- Good agreement locally in lift
- Different local trend in drag
- Up to ~ 4% discrepancy on integral loads

Rotor loads:

- More than 10% variation in thrust
- Up to ~ 20% deviations in torque, power consumption
- Up to 2.1° deviations in thrust tilt angle

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   - Mesh topology and solver setup
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   - Optimization environment
   - Twist optimization
   - Lift-based wing design
   - Alternative design
4. MODEL UNCERTAINTY ASSESSMENT
5. CONCLUSIONS AND FUTURE DEVELOPMENTS

Conclusions and Future Developments

05

Conclusions

- **Twist distribution** optimization **alone** is **ineffective** when wing lift is fixed
- **Redistributing** the **vertical force** burden toward the wing can yield efficiency **improvements**
- The **VDM** is **reliable** for wing **aerodynamic** predictions in cruise, but **rotor load** and **power** estimates exhibit **significant uncertainty**
- The alternative **dual-system** configuration shows **potential**, but due to the model uncertainty it can not be concluded whether this configuration truly increases power consumption or the VDM estimate is overly pessimistic

Future developments

- Experimental **validation**
- **Refine** the current **surrogate**-based results
- Investigate the **sweep** angle or the **taper** ratio
- Explore more **radical modifications**, inspired by the Cento and Pipistrel concepts
- Extend the **analysis** to **hover** and **transition** phases to provide a complete vehicle performance assessment

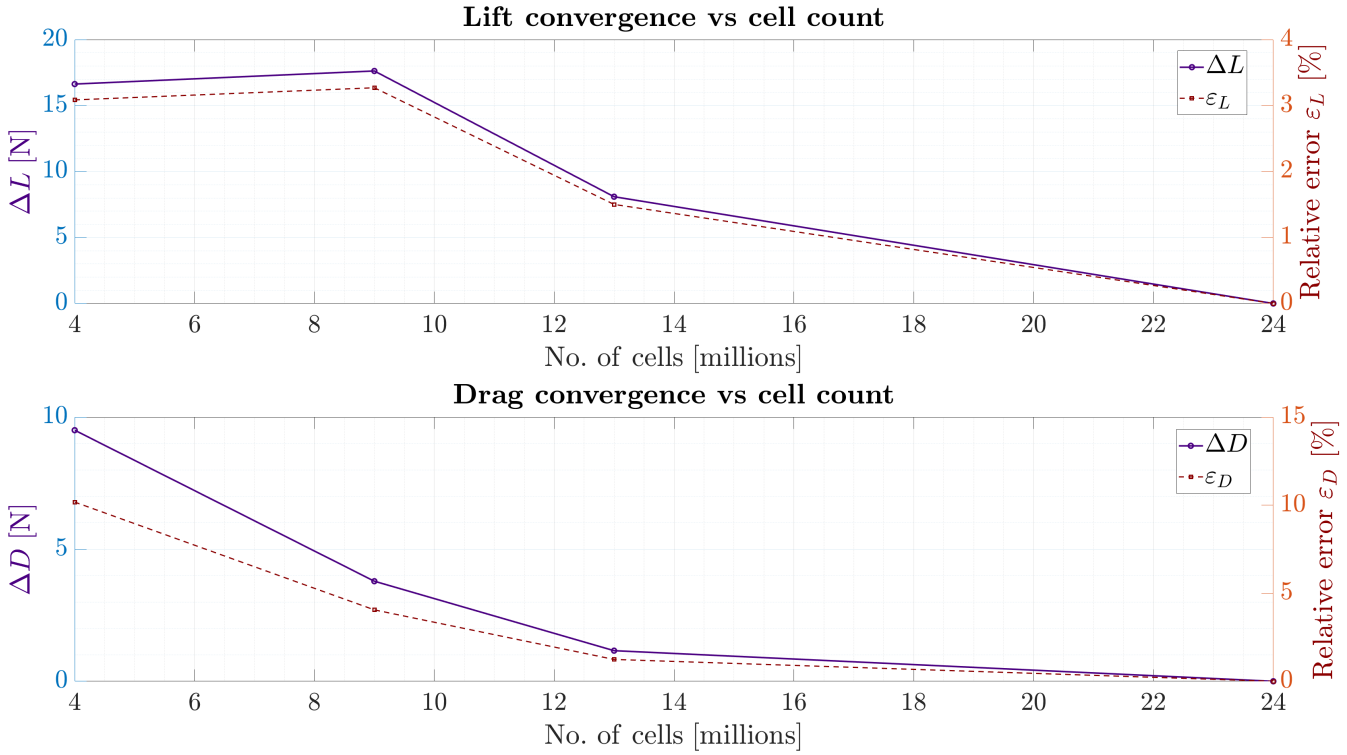
Aknowledgements

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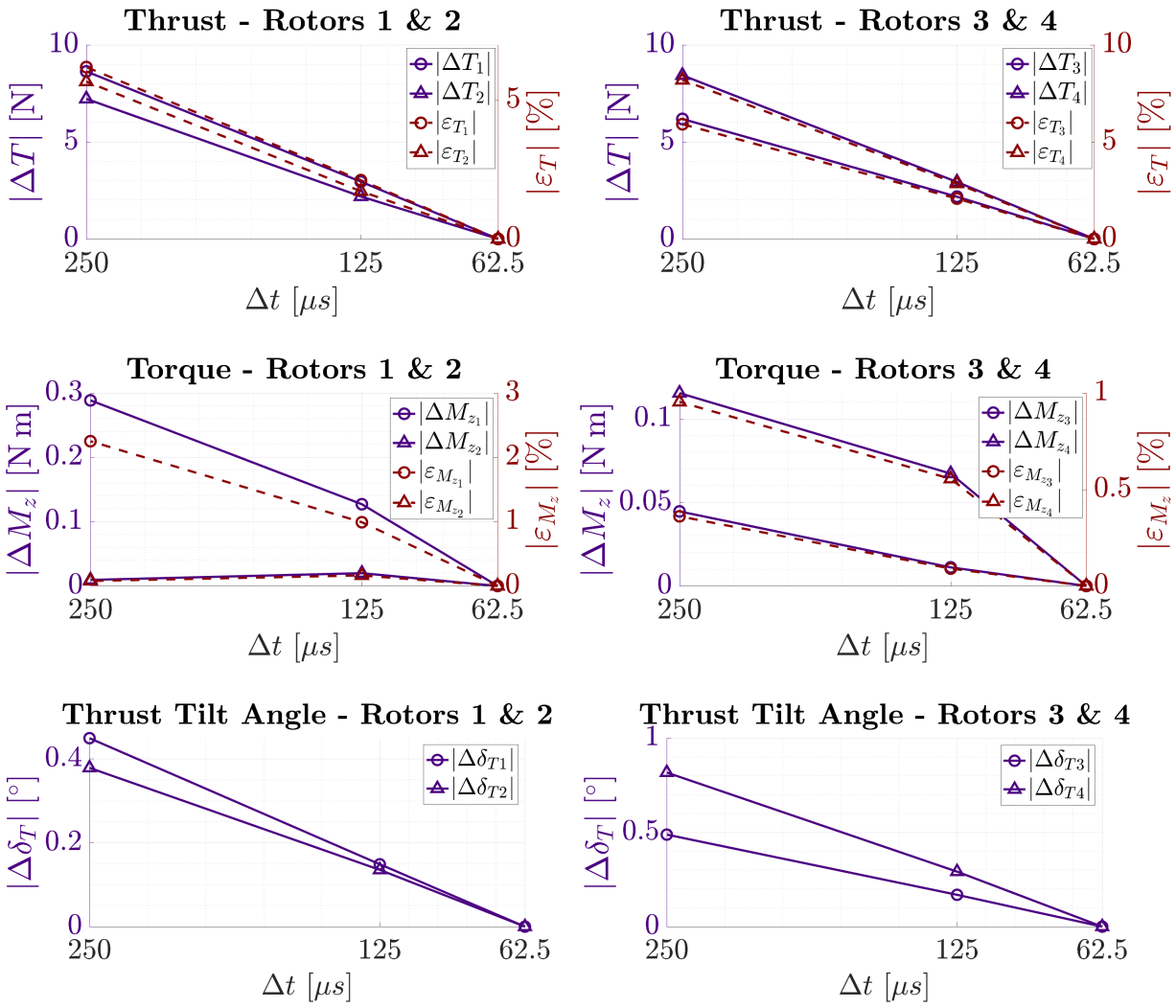
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URL https://www.pipistrel-aircraft.com/air-cargo

Validation – mesh independence

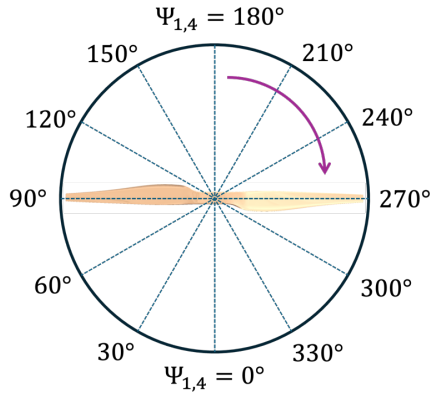


- VDM results
- 9-million-cell mesh
- Analogue study for the SMM mesh

Validation – time step independence



- $[2,1,0.5]^\circ/\Delta t \rightarrow \Delta t = [250,125,62.5]\ \mu s$
- $\Delta t = 125\mu s$ is selected



Alternative design – more results

Dual-system

Configuration	L^w [N]	D^w_{p+s} [N]	L^w/D^w_{p+s} [-]	L^{Tot} [N]	D^{Tot}_{p+s} [N]	$L^{\text{Tot}}/D^{\text{Tot}}_{p+s}$ [-]	$(T/P)_{\text{rotors}}$ [N/W]
Baseline	408.7	29.7	13.74	524.9	95.5	5.50	0.089
Zero-pitch alt.	434.3	25.3	17.15	620.7	103.8	5.98	0.184
$\boldsymbol{\Delta}_{\%}$	+6.3%	−14.7%	+24.8%	+18.3%	+8.7%	+8.7%	+107%

Configuration	T_{pusher} [N]	P_{rotors} [W]	P_{pusher} [W]		$P_{\text{rotors+pusher}}$ [W]	
			$\eta_p = 0.65$	$\eta_p = 0.8$	$\eta_p = 0.65$	$\eta_p = 0.8$
Baseline	/	5747	/	/	5747	5747
Zero-pitch alt.	96.3	2220	4446	3613	6667	5833
$\boldsymbol{\Delta}_{\%}$					+16.0%	+1.5%